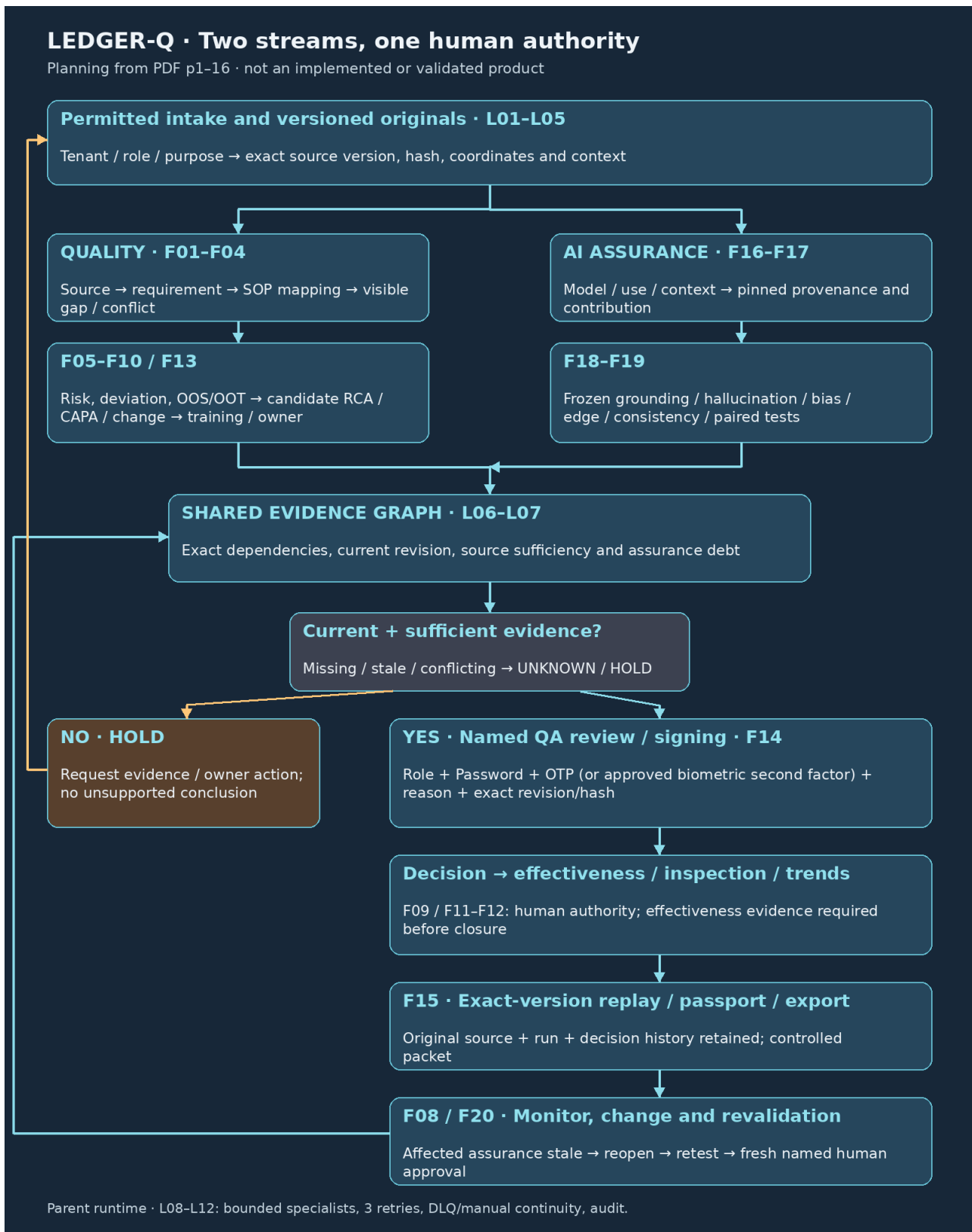


LEDGER-Q PDF + Video Mapping

6 October 2026 · Planning handoff · LEDGER-Q application not built in this task



Review scope and reference truth

The full 16-page PDF text and page layouts were inspected. All three complete recording timelines were sampled at 0.5-second cadence (2 frames/second), with detailed frames inspected for composition. This covers the observed sequence, not an assertion that every encoded frame was watched in real-time. These short clips show looping visual states; they do not show an end-to-end login, scrolling website, evidence graph, signature or export interaction. Those flows below are proposed tailored implementation behavior, not observed video behavior.

Ref	Exact supplied file	Duration	Actually visible	Allocation
V1	Screen Recording 2026-10-05 182302.mp4	6.549 s	Synchro light icy-blue framed hero; huge pale PRODUCTION word; center articulated robotic arm; compact top nav; small floating metrics/cards; orange CTA; cyclic joint/claw rotation	Public landing composition and slow hero object movement
V2	Screen Recording 2026-10-05 182549.mp4	16.400 s	Synchro six-tile branding board: moving arm card, DM Sans typography/color card, badge/ID card, logo notification tile, social-frame card and tilted device/app dock; largely stable collage with repeating arm motion	Public bento/proof tiles, design system and small contextual artifact cards; not a functional dashboard or full website journey
V3	Screen Recording 2026-10-05 182643.mp4	7.168 s	Factory Dashboard: dark slim left icon rail; pale blue canvas; title/search/user area; top-left moving arm + compact metrics; top-right violet/orange bars; bottom donut, system-log list and gradient card	Authenticated Command Center grid, navigation, log/queue composition and information hierarchy

Observed motion: V1 0–1.5 s angled open claw; about 2–4.5 s arm rotates/sweeps through frontal/left positions; by about 5–6 s returns toward earlier pose. V2 0–16.4 s preserves the branding-board layout while the top-left arm repeats. V3 0–4.5 s arm bends/rotates while charts/layout stay in place; about 5–7 s repeats toward the initial pose. No demonstrated scroll-trigger or page-to-page transition is inferred.

What will be kept, changed and added

Reference element	Keep	Tailor to LEDGER-Q	Reason / source
V1 framed icy-blue hero	Main composition, generous balanced space, nav and compact corner cards	LEDGER-Q title, quality/AI dual-stream explanation, real controlled-demo CTA; smaller 28–40 px statement instead of giant illegible heading	PDF p10 typography/readability

Reference element	Keep	Tailor to LEDGER-Q	Reason / source
V1 robotic articulated motion	Slow joint rhythm and subtle depth on public hero	Original evidence-stack / dual-stream mechanism: source cards and AI-use cards converge at a visible human gate; no industrial production claim	PDF p1/p6/p10 dual logic
V1 fake speed/24/7/100% counters	Placement only	Labelled saved-fixture counts/status or omit when no executed evidence; timestamps with completed-check semantics	PDF p14-15 claim discipline
V2 bento/card shapes	Mixed card sizes, thin borders, balanced gradients, compact label hierarchy	Source version tile; event chronology; signed decision specimen; release passport; adverse-test receipt; trace preview. Different relevant thumbnails per tile	PDF p9-10 actual product proof
V2 DM Sans and orange/violet branding	Font treatment and restrained accent motifs	Use licensed available font + verified fallback; navy/cyan active evidence; amber needs-review; red critical; green only approved	PDF p10 mandatory status semantics
V2 badge/social/device mockups	Framing pattern when useful	Version-bound review/permission card, public-safe work-sample and mobile review view; no invented staff credential, testimonial or client logo	PDF p8-10 roles and proof
V3 sidebar/top bar	Persistent rail, search/context, compact shell	Text-labelled canonical routes, tenant/product/site/use context, independent nav/content scroll, visible user/role/signout	PDF p9 enterprise shell
V3 robotic motion on operational dashboard	Spatial layout	Replace continuous decorative arm with quiet evidence/context card; limited functional selection/trace/state motion. Optional hero animation remains public-only	PDF p10 limits operational motion
V3 donut/bars/logs	Readable visual card composition	Evidence-debt categories; quality-event queue; critical AI failure denominators; overdue actions; registry release states; append-oriented real event log	PDF p9 command-center cards
Hover/scroll/close behavior not shown	No claim it exists in footage	Add keyboard focus, modest reveal on public page, dialog close/Escape/back, dependency-node selection; reduced-motion static alternative	PDF p15 accessibility + proposed UX addition

Do not copy screenshots or foreign logos/sample numbers into a working product. Recreate the composition with original LEDGER-Q components, source-bound data and lawful assets. “5D/7D/8D” is a visual request for depth and motion; implementation will be ordinary 3D/projection, not a technical capability claim.

Reference-based visual specification — proposed tailoring

Public page: cool blue/slate surrounding frame, soft blue/off-white inner cards, deep navy headings and restrained cyan evidence accents. Authenticated shell: deep navy rail/top context plus readable slate/light cards; event-status colors keep PDF meanings. Suggested starting tokens, not colors measured from original footage: navy #122338; slate #273E55; ice #E5EEF8; card #F5F7FB; cyan #2AB7D1; review amber #B67512; blocked red #B73642; approved green #197650. Verify contrast with actual text sizes before acceptance.

Hero: original dual-stream evidence mechanism moves slowly (12–18 s loop), with small depth/parallax and no fabricated live counters. Intro logo reveal, if used, ≤ 0.6 s and skippable; it must not block login or loading. Public hover elevation ≤ 3 px, zoom ≤ 1.015 ; scroll reveal 180–280 ms and repeatable only when readable. Dashboard motion is functional and brief: trace highlight, selected-node transition, status-confirmation. Respect reduced motion, hidden-tab/offscreen suspension and user pause. Loading/data errors remain textual and actionable, never masked by motion.

PDF page-by-page coverage

Pages	Requirements preserved here
1	Identity, dual quality/AI streams, human authority and design-target truth boundary
2	Architecture lock, F01–F10, proposed-control status, no standalone training OS
3	F11–F20 and shared L01–L07
4	L08–L12, P01–P05 and R01–R04
5	Permission-scoped deterministic/vision intake, fidelity coordinates, 3 retries/DLQ/manual continuity
6	Seven feature/layer linkage stages; quality/AI convergence; five release states
7	Signature gate, signing-time two-factor, reason, revision hash, failure states, Part 11 truth boundary
8	Eight user journeys, five user roles, command-center route
9	Remaining canonical navigation, eight command-center panels, desktop shell
10	Color/type/motion boundaries, eight public sections, reference stack
11	Remaining runtime stack, 16 core data objects, first API routes
12	Product/Site/User external UUID MDM readiness, mapping history and authorization separation
13	Remaining API routes, seven specialists, parent runtime/state/idempotency
14	Twelve test families and most outcome measures
15	Remaining metrics, security/privacy/recovery, specified→production maturity gates
16	Master build rules/order, usage optimization and final acceptance checklist

Canonical feature-to-layer-to-screen-to-proof map

Every row below is SPECIFIED / NOT BUILT IN THIS TASK. Proposed implementation allocations preserve PDF feature meanings. Full sortable version: [LEDGER_Q_Feature_Traceability.csv](#).

ID / canonical feature	PDF layers	Screen	Saved objects / owner	Must-pass evidence
F01 — Controlled Quality Source & Document Registry (p2)	L01-L04, L06	/sources	controlled_sources; Intake + evidence verifier	Retain original bytes, permission, effective date, predecessor and hash; reject unauthorized intake.
F02 — Requirement Extraction & Exact Evidence Traceability (p2)	L04-L07, L09-L11	/requirements	requirements; Requirement mapping specialist	Every accepted requirement links exact source/version/span; remove support and it must become UNKNOWN.
F03 — Regulation-to-SOP / Requirement Mapping (p2)	L04-L07, L09-L11	/requirements	sop_mappings; Requirement mapping specialist	Require context/version/rationale; missing SOP cannot be shown as covered.
F04 — Compliance Gap, Conflict & Evidence-Sufficiency Control (p2)	L05-L08, L10-L11	/requirements , /app	dependency_edges + evidence debt records; Evidence / audit verifier	Missing/conflicting/stale support remains visible debt and blocks unsafe progression.
F05 — Quality Risk, Severity & Confidence Assessment (p2)	L06, L08, L10-L11	/events , /review	quality_events + human_decisions; Domain specialist + QA	Separate severity, likelihood, controls, evidence sufficiency and confidence; no model score substitutes for human risk acceptance.
F06 — Deviation Investigation Review (p2)	L04-L09, L10-L11	/events	investigation_records; Deviation/OOS specialist	Preserve chronology, hypotheses and gaps; only named human determines root cause/disposition.
F07 — OOS/OOT Investigation Review (p2)	L04-L09, L10-L11	/events	quality_events + investigation_records; Deviation/OOS specialist	Retain laboratory source/method/result; ambiguous chromatogram requests review; AI never chooses disposition.
F08 — Bidirectional Change-Control & Impact Assessment (p2)	L04, L06-L12	/changes	dependency_edges; Change-impact specialist	SOP change stales dependent AI assurance; model/prompt/tool/data/policy change reopens affected quality/training review.
F09 — RCA-CAPA Lifecycle & Effectiveness Verification (p2)	L04, L06-L12	/capa	capa_actions; CAPA/effectiveness specialist	Separate proposed cause from human-established cause; closure blocked without implementation/effectiveness evidence.
F10 — Training Impact & Retraining Traceability (p2)	L02, L04, L06-L08, L10-L12	/training	training_impacts; Change-impact specialist + role owner	Link affected roles and completion evidence to correct changed version; no standalone training AI OS.
F11 — Audit / Inspection Finding & Readiness Management (p3)	L02, L04, L06-L08, L10-L12	/inspection	quality_events + capa_actions + exports; Evidence / audit verifier	Finding, gap, owner, action and closure reconstructable; unresolved debt stays visible.
F12 — PQR/APQR & Quality Trend Review (p3)	L04-L08, L11-L12	/trends	quality_events + evaluation_runs; Domain aggregation + human reviewer	Show denominators/time windows; correlation is never an automatic scientific conclusion.
F13 — Owner, Deadline, Escalation & Closure Control (p3)	L02, L08, L10-L12	/app , /capa , /review	capa_actions + assigned work; Parent workflow + owner	Persist due date/escalation; owner cannot use missing evidence to close work.

ID / canonical feature	PDF layers	Screen	Saved objects / owner	Must-pass evidence
F14 — Human QA Review, Override & Controlled Correction (p3)	L02, L04, L08, L10-L11	/review	human_decisions; Named authorized human only	Approve/revise/request/reject/suspend/escalate exact revision; stale signature or role fails closed; no silent self-training.
F15 — Audit/Version Reconstruction, Event Forensics & Controlled Export (p3)	L02-L04, L07-L12	/audit	audit_events + exports + release_passports; Evidence / audit verifier	Reconstruct original versions and AI contribution; export permission-filtered approved evidence; preserve open gaps.
F16 — AI Model / Use / Context & Release Registry (p3)	L01-L04, L06-L08, L10-L11	/ai-registry	ai_uses + release_passports; AI provenance specialist	Intended/excluded use, owner, reviewer and exact model dependencies required; no unsupported VALID state.
F17 — AI Provenance, Transparency & Contribution Record (p3)	L03-L07, L09-L11	/ai-registry , /audit	ai_runs; AI provenance specialist	Retain prompt/config/tools/inputs/retrieval/output/model/human use; immutable past run identity.
F18 — AI Evaluation, Grounding & Hallucination Suite (p3)	L05-L07, L09-L11	/ai-evaluation	evaluation_runs; AI evaluation specialist	Frozen expected/actual evidence and citation checks; conflicting assessors and critical failures remain visible.
F19 — Bias, Edge-Case & Consistency Verification (p3)	L04, L06, L09-L11	/ai-evaluation	evaluation_runs; AI evaluation specialist	Subgroups, adverse cases, repeated-run variation and paired model comparisons with denominators/limits.
F20 — Assurance Monitoring, Regression, Blast Radius & Revalidation (p3)	L04, L06-L12	/changes , /ai-evaluation , /app	evaluation_runs + dependency_edges + release_passports; Change-impact + AI evaluation specialists	Changed dependency invalidates current assurance; retest plus fresh human approval needed before restored use.

Shared layers — no parallel replacement architecture

ID	Exact PDF name	Implementation responsibility
L01	Workspace & Tenant Isolation	Tenant filters at database, object storage, retrieval, graph, tasks and exports.
L02	Identity, Roles & Approval	Named QA, owner, SME, admin and approver with current authority checks.
L03	Authorized Source & Intake	Permission, purpose, type, byte limit and approved-source checks before parsing/provider calls.
L04	Storage, Metadata, Versioning & Release Records	Originals, hashes, histories, model/config versions, runs and release passports.
L05	Domain Parsing & Normalization	Typed text/tables first; vision only for visual evidence; retain original coordinates.
L06	Source / Policy / Context & Staleness Governance	Authority/intended-use/policy/context and bidirectional invalidation rules.
L07	Evidence Graph, Retrieval, Assurance Debt & Blast Radius	Relational edges linking requirements, quality objects, AI uses, decisions and outputs.
L08	Workflow State, Reopening & Orchestration	One parent controls routing, waits, leases, retries, reopening and safe terminal states.
L09	Multi-Agent & Multi-Model Routing	Bounded named candidate specialists with model pins and outage protocol.

ID	Exact PDF name	Implementation responsibility
L10	Safety, Security & Human Decision	No autonomous regulated root cause/disposition/closure/release; enforce human authority.
L11	Evaluation, Audit, Correction Firewall & Revalidation	Frozen tests, corrections, failure preservation, regression, retest and traceable evidence.
L12	Integration, Operations & Scale	Reviewed APIs, notification outbox, worker operations, backups, rollback and later scale.

Proposed child controls and challenge controls

These are retained from the PDF, not verified world-first claims. Their feature/layer allocations below are proposed implementation links.

ID	PDF control	Attached feature/layer	Proof required
LQ-P01	Cross-domain dependency invalidation	F08/F20; L06-L08/L11	Test quality→AI and AI→quality invalidation without changing historical approvals.
LQ-P02	Reconstructable failure / AI-contribution forensics	F06/F07/F15/F17; L04/L07/L11	Replay exact original contribution and first failure; unresolved cause remains UNKNOWN.
LQ-P03	Human-correction regression firewall	F14/F18/F20; L04/L10/L11	Correction → proposed regression case → QA approval → version → retest; no silent training.
LQ-P04	Assurance debt + conflicting-assessor / blast-radius panel	F04/F08/F18/F20; L07/L11	Deterministic/model/human disagreements and affected objects visible; missing evidence cannot be outvoted.
LQ-P05	Non-gaming trust report	F12/F18-F20; L11	Task success, abstention, critical failure cost, review burden and denominators shown separately.
LQ-R01	Pharma critical-error challenge pack	F18/F19; L11	Mutate units, operators, batch, version and evidence; retain actual failed cases.
LQ-R02	Evidence-removal challenge	F04/F18; L07/L11	Remove support; demand abstention or a valid authorized alternate source.
LQ-R03	Task-specific model-change comparison	F19/F20; L09/L11	Paired critical cases stay visible even if aggregate score improves.
LQ-R04	Failure-location report	F15/F18/F20; L11	Report first evidenced break; do not fabricate the unresolved cause.

PDF enhancements requiring explicit implementation

Vision intake (p5): L01/L03 check tenant/source/purpose permission; L05 runs deterministic extraction first and conditionally routes chromatograms, HPLC graphs, handwritten notes and nested tables to approved vision/document AI. Candidate JSON retains source version, page/region, table/series ID, raw observation, units, method, confidence and parser/model identity. Original remains authoritative.

Unreadable/ambiguous → UNKNOWN/NEEDS_REVIEW. “Loss-minimizing” target, not unvalidated “lossless” claim.

Outage handling (p5): L09 eligible AI/API failure → three automatic retries with exponential backoff → DLQ for noncritical tasks. Persist exact revision/run/cause/attempt. Recovery checks current authorization/policy. Critical quality event, evidence, rules, owner routing and named QA work stay available in deterministic/manual mode. AI outage never suppresses a quality state or closes/releases work.

Signing (p7): F13/F14 configured decision transition → exact-revision human review → SIGNATURE_REQUIRED → Password + OTP or approved biometric second factor at signing → role/current revision/hash/reason check → approved/rejected event and UTC audit record. Reason choice Authored/Reviewed/Approved; append-oriented revision/decision hash with chain reference where implemented. Failure → HOLD/REAUTHENTICATE/RE-REVIEW. A biometric route requires an approved real identity method; it is not simulated by an icon. Compliance-targeted, not proven Part 11 compliance.

MDM (p12): Product_ID, Site_ID and User_ID remain stable local keys with optional external_uuid, external_system, mapping_status, effective time and mapping history.

UNMAPPED/MATCHED/CONFLICT/RETIRED. An external identity match does not grant tenant access, reviewer authority or signature rights. Only authorized mapping can contribute to cross-module blast radius. LEDGER-Q is not the enterprise master-data authority.

Nine agentic parts — allocated without replacing L01-L12

Part	Where	Actual responsibility / proof
Goal interpretation / planning	L08 + L09	Bounded task plan, required evidence and next permitted action; failure/conflict may replan, never bypass human gate
Tool integration	L03/L05/L12	Approved source/parsing/DB/provider/export adapters; allowlist/permission/typed outputs tested
Memory	L04/L06/L07	Current approved context + versioned history; tenant-scoped retrieval; corrections do not silently become memory
Orchestration / runtime	L08/L12	One parent, persisted jobs, waits/retries/leases/idempotency/DLQ; works without browser
Multi-agent collaboration	L09 + L11	Seven scoped specialists, shared exact revision, candidate merge/disagreement; no approval by voting
HITL	L02/L08/L10	Named reviewer exact revision, revise/request/reject/suspend/escalate; signing gate
Safety / policy	L01-L03/L06/L10	Tenant/role/purpose, injection defense, network/file limits and blocked unauthorized effects
Observability / evaluation	L11/L12	Correlation trace, frozen expected/actual tests, latency/cost when available, retained failure and denominators

Part	Where	Actual responsibility / proof
Governance / accountability	L04/L06/L10/L11	Provenance, append history, controlled correction, retention and version-bound passport/export

“Agentic” is justified only by executed planning/routing/collaboration records. Deterministic specialist functions are labelled deterministic. A live provider receipt is separate from scientific validation. No decorative unrestricted swarm.

Shared data and agent handoff contract — proposed implementation detail

Every candidate packet carries tenant/workspace, correlation/run, task kind, source ID/version/hash/span, input/output revision, policy/config/model version, permitted purpose, gaps/uncertainties, status, owner and next action. Original records live in the controlled store; agents pass authorized references and typed candidates, not a global unscoped memory dump.

1. Parent checks authority, exact revision, purpose and idempotency; freezes input fingerprint.
2. It routes only needed tasks to seven named specialists and waits for bounded outcomes.
3. Specialists cannot grant roles, pick arbitrary external tools or approve their own output. A document/prompt injection is data, not an instruction.
4. Evidence verifier and deterministic schema/numeric/source checks compare candidates. Disagreement returns visible debt/UNKNOWN/HOLD, not majority acceptance.
5. Parent commits candidates with compare-and-swap on the current revision and stores provenance. Late/stale responses are retained as failed/cancelled traces without overwriting new work.
6. Human decision binds exact revision and source/test dependencies; changed evidence invalidates current approval and requires new review/signature.

Specialist (PDF p13)	Reads / produces	Prohibited authority
Requirement mapping	Controlled sources → requirement/span/mapping candidates	Invent coverage / approve compliance
Deviation/OOS	Quality originals + chronology → gaps and hypotheses	Determine root cause/disposition
Change impact	Dependency edges + exact changes → blast-radius/stale candidates	Close/release affected controls
CAPA/effectiveness	Approved cause/action/evidence → effectiveness candidates	Approve CAPA closure
AI provenance	Pinned model/prompt/tool/input/output → contribution record	Rewrite prior run / hide config
AI evaluation	Frozen approved cases → grounding/bias/edge/regression outcomes	Hide critical failures in aggregate
Evidence/audit verifier	Version hashes + decision chain → integrity/reconstruction checks	Override named human QA authority

Core saved objects and relationships (all 16 from PDF p11)

Object	What must be saved / connection
workspaces	Tenant, site/product/process context, policy/schema/workflow version
controlled_sources	SOP/policy/form/record/dataset, permission/owner/effective date, version/hash/predecessor
requirements	Typed requirement, exact source/span/version and applicability rationale
sop_mappings	Requirement → SOP/process/version; coverage, gap/conflict state
quality_events	Deviation/OOS/OOT/finding/change, chronology/risk/status/evidence
investigation_records	Hypotheses, gaps and evidence; human disposition and AI-contribution refs
capa_actions	Human-established/candidate cause distinction, actions, owner/due/implementation/effectiveness/closure
training_impacts	Changed version, affected role, requirement and completion evidence
ai_uses	Model/version/use/exclusions/context/dependencies/owner/reviewer/release state
ai_runs	Exact prompt/config/tools/input/retrieval/output/model identity/human use
evaluation_runs	Frozen case/version, expected/actual, critical failures, grounding/bias/edge/first break
dependency_edges	Source/SOP/process/product/batch/training/AI/output/decision links with stale state
human_decisions	Exact revision/action/rationale/residual risk/approver/UTC/signing metadata
release_passports	Use + context/version/evidence/tests/limits/approver/state/history
audit_events	Ordered cross-stream events with correlation/revision/hash references
exports	Controlled packet manifest, redactions, exact versions and release flag

Proposed supporting tables (not extra canonical features): persisted jobs/checkpoints/DLQ, notification outbox, immutable source spans, evidence-debt items, external mapping history, signature factor events and approved regression-case versions. Needed to implement the PDF behavior transparently; exact schema chosen at stack lock.

API contract allocation (all 12 proposed PDF endpoints)

Route	Purpose	Guard
POST /sources	Register source/version	Permission/type/purpose/tenant
POST /requirements/extract	Candidate requirements/spans	No accepted requirement without exact evidence
POST /mappings	SOP/process/version mapping	Missing rationale → review

Route	Purpose	Guard
POST /events	Quality event + chronology	Typed source context; no AI disposition
POST /impact/calculate	Bidirectional affected objects	Mark stale/reopen; preserve history
POST /capa	Action/effectiveness record	Evidence + human closure
POST /ai-uses	Intended use/model/context	No unsupported VALID state
POST /ai-runs	AI contribution/provenance	Immutable exact run/evidence identity
POST /evaluations/run	Frozen/adverse case results	Denominators/first failures retained
POST /reviews/{id}/decision	Named QA action	Current role, expected revision, signature reason/factors
POST /revalidate	Critical retest after change	Fresh review required before release state update
POST /exports	Controlled evidence/passport packet	Permission-filtered, version-bound approved evidence

The PDF lists 12 endpoints (3 on p11 and 9 on p13). This count is checked against the rows above. Read-only GET detail/queues/history and explicit authentication/signature-step endpoints are proposed supporting APIs, not new top-level features. Worker resume/cancel/dead-letter endpoints require authorization and idempotency.

Exact workflow and release-state semantics

Parent state sequence from PDF p13: **START** → **AUTHORIZED** → **INPUT_VERSIONED** → **PARSED** → **CANDIDATE_READY** → **VALIDATED** → **HUMAN_REVIEW_REQUIRED** → **APPROVED** / **REVISED** / **REQUEST_EVIDENCE** / **REJECTED** → **ACTION_OPEN** → **VERIFIED** / **CLOSED**, with visible **HOLD** / **EXHAUSTED** branches. These are workflow states, not automatic regulatory disposition. No failed step becomes success.

Signature sequence from PDF p7: **DRAFT** → **HUMAN_REVIEW_REQUIRED** → **SIGNATURE_REQUIRED** → **AUTHENTICATED** → **APPROVED** / **REJECTED**. Current role, two-factor result, controlled reason and exact revision/hash must pass at the signing event. Authentication failure, stale revision, missing reason or mismatch means **HOLD** / **REAUTHENTICATE** / **RE-REVIEW**.

AI-use release state (PDF p6)	Exact meaning
VALID	Approved evidence and intended-use context remain current
SUSPENDED	Temporary hold while an issue or change is investigated
EXPIRED	Time, context or version trigger requires revalidation
REVOKED	Evidence/control failure makes prior release unusable
REVALIDATED	Required retest plus named human approval restore controlled use

This state applies to the defined AI use/passport; it is not batch-release authority, medicine approval or an autonomous quality-event closure. Old states remain reconstructable after any transition.

Canonical navigation and video links

Route	Purpose	Reference + data
/	Public landing	V1 hero, V2 bento, PDF p10 eight sections
/app	Assurance Command Center	V3 grid; saved open events/debt/release states/overdue/critical tests
/sources	Controlled Sources	Quiet tables/document preview; F01 originals
/requirements	Requirements & SOP Map	Mapping table + exact source drawer; F02-F04
/events	Quality Events	Chronology/evidence/risk workbench; F05-F07
/changes	Change & Blast Radius	Functional node/edge highlight on selected saved dependency; F08/F20
/capa	RCA-CAPA & Effectiveness	Owned action/evidence/closure; F09/F13
/training	Training Impact	Correct-version role/record list; F10
/inspection	Inspection Readiness	Finding→action→evidence packets; F11/F15
/trends	PQR/APQR & Trends	Saved denominators/time windows, exceptions; F12
/ai-registry	AI Use & Release Registry	Use/model/context/passport/state; F16/F17
/ai-evaluation	AI Evaluation Lab	Frozen cases/actual failures/paired comparison; F18/F19
/review	Human QA Review Queue	Exact evidence/revision + signing workflow; F14
/audit	Replay / Forensics / Export	Immutable history/source/run/decision reconstruction; F15
/settings	Organisation / Controls	Role/policy/test pack/integration/notification settings; L02/L06/L12

Detail panels always show status, source/version, owner, uncertainty, limitation and next permitted action. Keep nav independently scrollable and content panel stable. A graph panel is proposed LEDGER-Q functionality from the PDF, not an interaction shown in these videos.

Public landing sequence

1. Optional short logo reveal → V1-framed LEDGER-Q hero with dual-stream mechanism and controlled-demo CTA.
2. Problem strip → fragmented quality evidence and the separate AI trust problem.
3. How it works → quality + AI evidence converge on named human authority, saved versions and reconstructable history.
4. Evidence/human control → missing/stale/conflict examples and exact-source preview.
5. V2-inspired product bento → different source/event/review/passport/trend/trace thumbnails from actual synthetic records once built.
6. Architecture → source/API/parent/state/evidence/evaluation/human gate; restrained layered depth.
7. Proof/limitations → executed receipts only, scope/date/denominator; known gaps visible.

8. Closing CTA/contact → actual demo/work sample/code links only when available. No fake waitlist position or inactive link.

Visitor → workspace connection flow

Editable branched flowchart is supplied in the Markdown and matching .mmd file. The overview diagram is included above.

Public case must not create a customer workspace or imitate professional approval. Sign-in and document permission prompts differ: public safe read needs no fake authentication; protected work and provider transmission use their actual authorization/consent.

Master backend flowchart — two streams, one authority

Editable branched flowchart is supplied in the Markdown and matching .mmd file. The overview diagram is included above.

F14 governs regulated decisions across F05–F20. A permitted AI-use passport state is not permission to release a medicine or autonomously close a quality event.

Runtime and provider failure connection

Editable branched flowchart is supplied in the Markdown and matching .mmd file. The overview diagram is included above.

Manual continuity does not fabricate a failed AI task’s result; it preserves critical quality operations and the unresolved provider gap.

Synthetic integrated work sample — proposed fixture, not built

Product: SQ-100 immediate-release tablet, Site SYN-S1, Batch SYN-B01; wholly synthetic portfolio exercise. No commercial disposition or regulatory submission.

Episode	Fixture	Expected correct behavior
Requirement/SOP	Synthetic source requires current documented assay procedure; SOP-QA-101 v1 covers it	Exact requirement span + reviewed mapping; no source-free coverage
Deviation/OOS candidate	Lab record A and B report incompatible assay entries for same batch/method	Conflict visible; no selected/averaged authoritative result and no autonomous OOS disposition
Visual intake	Labelled synthetic chromatogram with an ambiguous area/handwritten annotation	Original image retained; conditional vision candidate + coordinates; ambiguity UNKNOWN
CAPA candidate	Investigator proposes transcription-check action	Candidate cause/action only; named QA decides; effectiveness evidence still required
Training change	SOP-QA-101 v2 changes approved handling steps	Affected roles/completions for v1 become stale; new-version training/review trace
AI onboarding	Lab-summary AI use, intended-use exclusions, model/prompt/config pins	Frozen source-grounding tests; unsupported answer escapes blocked; no VALID before review

Episode	Fixture	Expected correct behavior
Quality → AI change	SOP v2 changes required evidence	Dependent AI tests/passport suspended/expired as policy requires; retest/review
AI → quality change	Model/prompt/tool revision changes behavior	Affected SOP/validation/training/quality controls reopened with graph path
Correction firewall	Human corrects model value	New governed correction; benchmark/memory unchanged until authorized regression approval/version
Historical replay	Select earlier decision before change	Reconstruct then-current source/run/decision; current versions do not rewrite history
Signing negative	Wrong role, expired OTP, missing reason or stale revision	HOLD/reauth/re-review; no signature completion
Controlled positive path	All required fixture evidence supplied, tests current, named test-human authority exercised	Version-bound recorded decision; still labelled synthetic exercise, not real professional qualification

A second intentionally unresolved case stays HOLD on missing effectiveness evidence. Demonstrate both successful bounded workflow and preserved uncertainty; do not invent closure to make the demo look complete.

Test families — all 12 from PDF p14

Family	Must demonstrate
Source / requirement	Authorized vs wrong purpose; exact span; stale version
Mapping / gap	Covered requirement vs missing SOP/conflicting versions
Deviation / OOS	Chronology completeness; absent evidence; AI hypotheses vs human disposition
Change / blast radius	Both quality→AI and AI→quality; unrelated objects remain current
CAPA / effectiveness	Implemented action vs missing effect proof/recurrence; closure block
Human correction firewall	Governed correction and blocked silent benchmark/memory update
AI grounding	Supported citation vs missing/unsupported claim
Bias / edge / consistency	Subgroups, repeated runs, paired critical failures
Release passport	Valid→change→suspended/expired→retest→human revalidated
Assurance debt	Missing/stale/conflicting evidence + affected object paths
Tenant / role	Wrong tenant/role retrieval/review/export denied
Restore / replay	Historical quality + AI decision restored with matching original IDs/hashes

Additional implementation checks: provider redirects/allowlist/private-network rejection, retries/DLQ replay/idempotency, late responses/concurrency, signature factor/reason/hash, malformed files/parsing limits, injection attempts, public safe projection, keyboard/dialog/focus, nav scroll isolation, reduced motion and mobile.

Outcome measurements — all 12 PDF measures, not invented results

Measure	Denominator / interpretation
Requirement trace coverage	Accepted requirements with exact source/version/location ÷ accepted requirements
Open assurance debt	Count + age of missing/stale/conflicting items by severity
Quality event review completeness	Events with required chronology/evidence/reviewer fields ÷ reviewed events
CAPA effectiveness completion	Due CAPAs with required effectiveness evidence + human closure ÷ CAPAs due
Reopened dependency precision	Correctly reopened objects ÷ reopened objects in frozen change cases
AI grounding pass rate	Supported/cited expected cases passed ÷ grounding cases; critical failures separate
Unsupported output escape rate	Unsupported outputs incorrectly accepted ÷ unsupported challenge cases
Abstention / request-evidence rate	Abstained/evidence-request cases ÷ evaluation cases; show task success alongside
Critical edge-case pass	Critical cases passed ÷ critical cases; no aggregate masking
Release-state freshness	VALID uses with current required dependencies/tests ÷ all VALID uses
Human override/correction rate	Reviewer overrides/corrections ÷ reviewed AI-assisted outputs; rationale categories
Inspection reconstruction completeness	Required source/event/action/decision items recovered ÷ required reconstruction items

A 0 denominator is NOT_MEASURED/NOT_APPLICABLE with reason, not automatic 100%. Real active-effort savings and error reduction require qualified expected answers, users and a comparable manual baseline; not demonstrated by fixtures.

Proposed gap-closing additions — no extra canonical features

Proposal	Existing home	Why / proof
A01 Version-locked shared contract registry	L04/L08/L11	Single schemas and source/model/policy fingerprints; reject mismatched candidates
A02 Independent nav scroll + responsive workbench	L12 / frontend	Prevent sidebar moving with detail content; tested keyboard/mobile paths
A03 Public saved proof projection	F15/L01/L10	Read-only synthetic canonical record, hash/readback verified; no private workspace or approval impersonation
A04 Decision-ready detail drawer	F14/L07	Always status/evidence/version/owner/limit/next action; close/Escape/back/focus tests

Proposal	Existing home	Why / proof
A05 Privacy-safe public assurance summary	F18-F20/L11	Expose scope/outcome/limitations only; never prompts/secrets/private controls
A06 Operator heartbeat + completed-run freshness	L08/L12/F20	Scheduled config separate from actual completions; failed/overdue → evidence gap
A07 Export manifest with expiry/verification state	F15/L04/L10	Current hash/revision/dependencies/authorization; stale packet cannot masquerade as current release
A08 Risk-based work priorities	F05/F13/L08	Owner can inspect due/critical/open debt; no hidden auto-disposition

These are proposed implementation details needed to close common BEACON/FORGE gaps. They are not new verified product differentiators, not extra F21+/L13+ and not already deployed.

Security and operational rules preserved

Least privilege and tenant/role checks before retrieval/review/export/model access; network allowlist and redirect recheck/private-network block/timeout/size control; current-revision concurrency; server-side secrets; append-oriented auditable history without unsupported tamper-proof claim; original-ID/hash restore rehearsal; confidential data only under appropriate permission/contracts/environment controls; accessible contrast/focus/labels/reduced motion. Parent worker independent of browser; measured latency/cost reported only when actually captured. Scale according to tested workload, not “global enterprise” marketing.

Next action after review

First freeze stack + schemas/state + synthetic expected outcomes; then source→requirement→gap→event→human decision saved slice, followed by connected AI assurance/change/revalidation and controlled export. This task delivers the plan and mappings. LEDGER-Q app, credentials, deployment, professional validation and production outcomes are not claimed complete.